PROJECT REPORT

Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau

TEAM ID: LTVIP2026TMIDS81059

PRESENTED BY:

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1. INTRODUCTION

1.1 Project Overview

This project involves the development of a **Comprehensive House Data Analysis** system. The primary focus is to analyze multi-dimensional housing datasets to identify the core drivers of property pricing. By integrating factors such as house age, renovation history, and structural features, the project provides a unified visual platform for strategic real estate decision-making.

1.2 Purpose

The purpose is to solve complex problems in a way that fits the state of the customers, such as real estate analysts and marketing managers. It aims to simplify complex and scattered housing data into actionable insights to reduce market confusion.

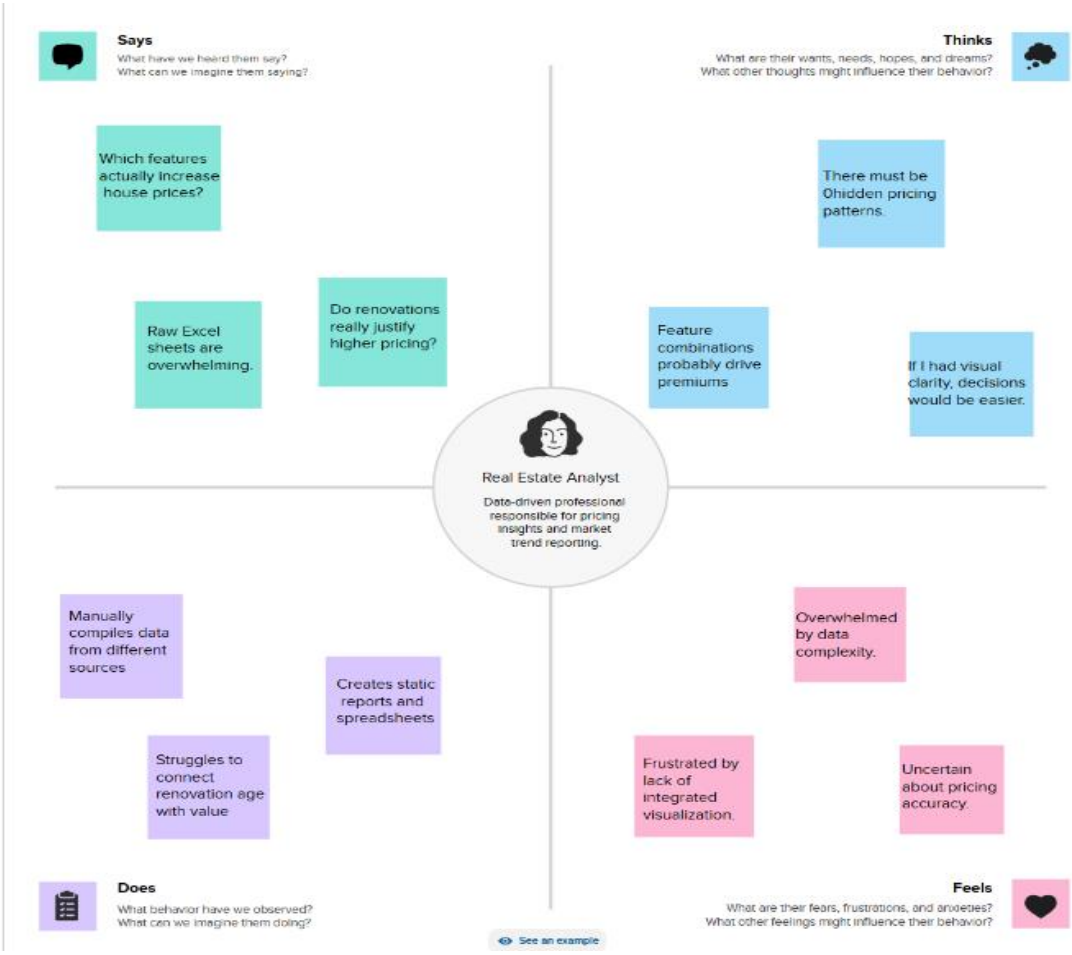
2. IDEATION PHASE

2.1 Problem Statement

- **Core Issue**: Stakeholders lack a consolidated visualization system to connect house age, renovation history, and features with prices.
- **Analyst Perspective**: Data is multi-dimensional and difficult to interpret in raw tabular form.
- **Marketing Perspective**: Lack of visibility into how renovations and features impact competitive positioning.

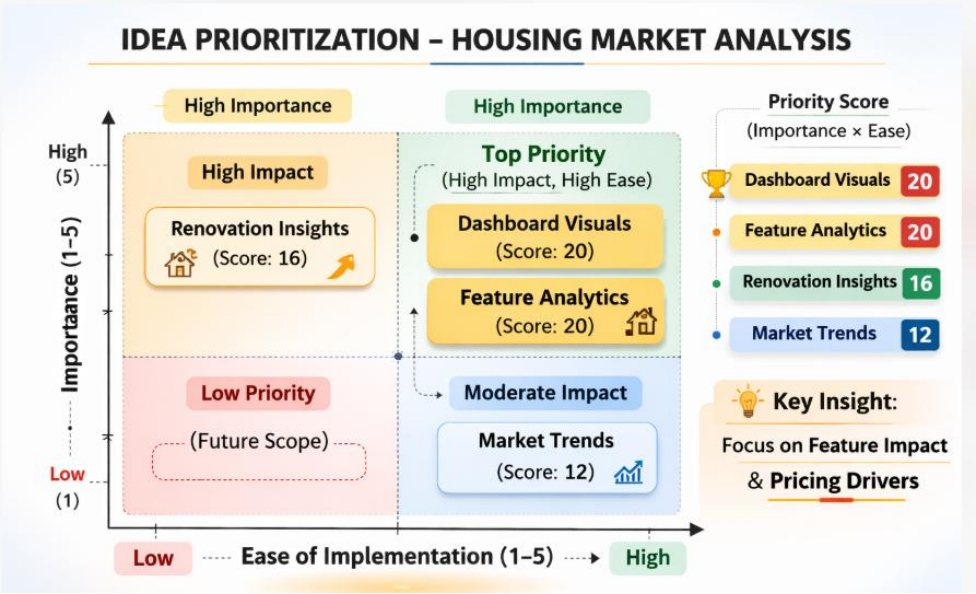
2.2 Empathy Map Canvas

- **Says**: "Which features actually increase house prices?".
- **Thinks**: "There must be hidden pricing patterns".
- **Does**: Manually compiles data from different sources and creates static reports.
- **Feels**: Overwhelmed by data complexity and uncertain about pricing accuracy.



2.3 Brainstorming

- **Prioritization**: "Dashboard Visuals" and "Feature Analytics" were identified as top priorities with a Priority Score of **20**.
- **Top Priority**: Build an interactive dashboard integrating features, renovation history, and sale price trends



3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

- **Stages**: Entice (Understand trends), Enter (Access dashboard), Engage (Apply filters), Exit (Export insights), and Extend (Future strategy).

CUSTOMER JOURNEY MAP

Project: Housing Market Trends & Pricing Analysis Dashboard

Entice	Entice	Enter	Engage	Exit	Extend
Help me...	Understand housing price trends & feature impact	Access dashboard & explore housing data	Analyze renovation impact, features, and pricing patterns	Export/share pricing insights	Use insights for future strategy & forecasting
Goals & Motivations	Identify price drivers	View interactive charts & summaries	Apply filters (age, bedrooms, bathrooms, floors)	Generate reports for decision-making	Improve pricing strategy & competitive positioning
Areas of Opportunity	Market confusion about pricing factors	Data complexity & overload	Need deeper feature correlation insights	Simplify export & summary views	Add predictive analytics & trend forecasting
Interactions	Company website / presentation	Tableau dashboard homepage	Interactive charts, filters, histograms	Export PDF / report sharing	Follow-up analysis meetings
Things: What digital touchpoints or physical objects would they use?	Laptop / desktop	Dashboard interface	Filters, bar charts, histograms, pie charts	Export buttons, report files	Saved dashboards, updated datasets
People: Who do they talk to?	Marketing team	Data analyst	Executive team	Stakeholders	Investors / Strategy team
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3.2 Solution Requirements

- **Functional (FR)**: Data integration (.csv), dynamic filtering (age, bedrooms, floors), and pricing analytics.
- **Non-Functional (NFR)**: Dashboard load time < 3 seconds and secure authenticated login.

+1

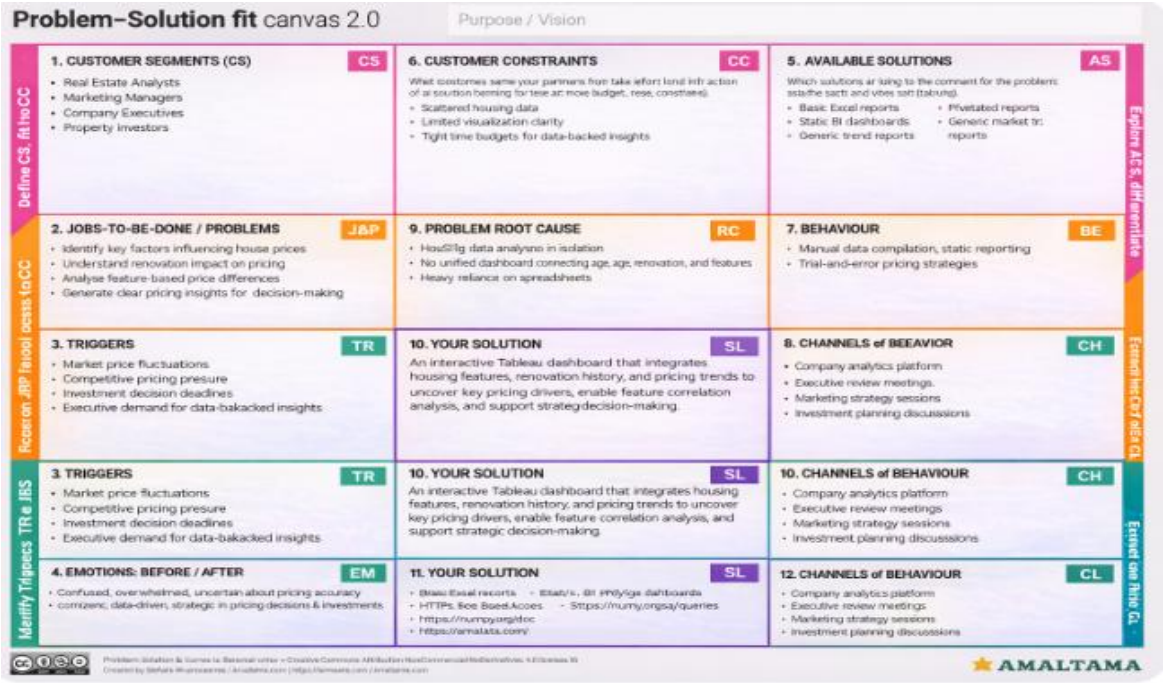
3.4 Technology Stack

- **Visualization**: Tableau Public / Desktop.
- **Processing**: Python (Pandas, NumPy).
- **Database**: PostgreSQL / MySQL.
- **Infrastructure**: AWS S3 for storage and AWS EC2 for hosting

4. PROJECT DESIGN

4.1 Problem Solution Fit

- **The Solution**: An interactive Tableau dashboard that integrates housing features and pricing trends to support strategic decision-making.

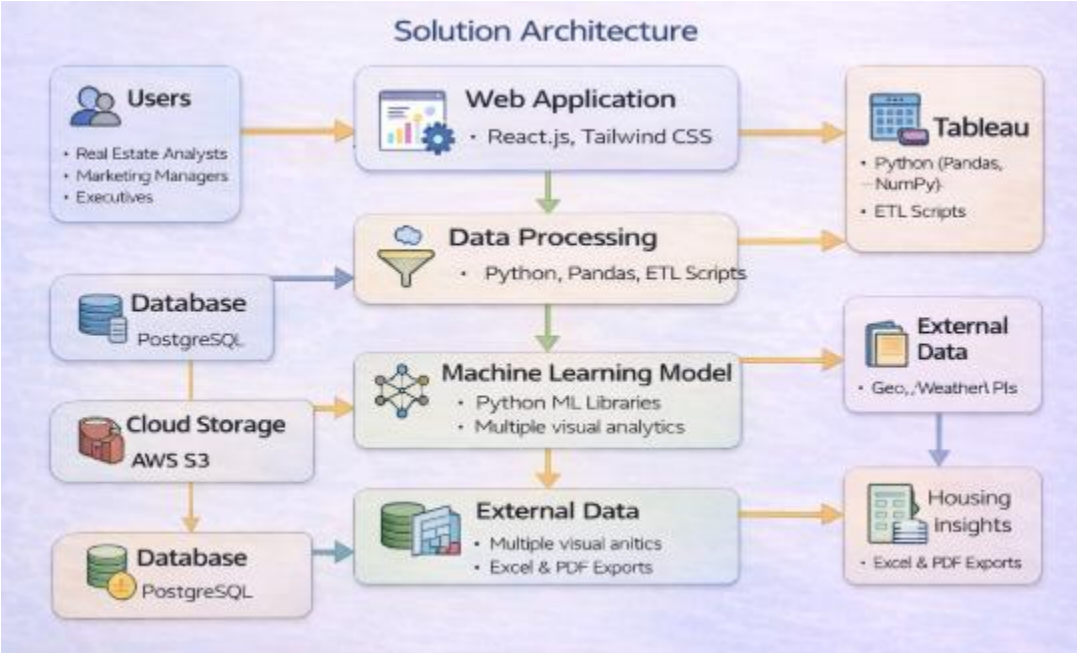


4.2 Proposed Solution

- • **Problem Statement:** Stakeholders lack a unified, data-driven way to see how house age, renovations, and property features dictate market value, leading to **inefficient pricing and "gut-feel" decisions.**
- • **Proposed Solution:** An **interactive Tableau ecosystem** that integrates disparate housing data into a single source of truth for analyzing feature correlations and pricing trends.
- • **Novelty & Uniqueness:** Shifts from static reporting to **dynamic, story-based insights** where users can filter by age, renovation status, and layout in a single integrated view.
- • **Social & Customer Impact:** Eliminates pricing bias and builds **investor confidence** through transparent, data-backed strategies for more reliable property appraisals.
- • **Business Model:** Potential for **B2B licensing (SaaS)**, enterprise BI consulting, or a subscription-based model for real estate analytics.
- • **Scalability:** Designed to ingest national-level datasets and evolve into a **predictive forecasting tool** using machine learning for future market trends.

4.3 Solution Architecture

- **Workflow**: Users interact with a Web Application (React.js) → Data Processing (Python) → Machine Learning Models → Data Analysis (Tableau).



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Sprint 1: Foundation & Data Preparation

Duration: 1 Feb – 7 Feb 2026 (7 Days) | **Total Story Points:** 15

Task ID	User Story / Task	Priority	Assigned To	Points
USN-1	Finalized housing pricing problem & scope	High	Sai Charan	2
USN-2	Identified housing dataset attributes	High	Rammya Sre	3
USN-3	Collected & reviewed housing dataset	High	Chandrika	3
USN-4	Cleaned missing renovation & feature values	High	Bhoomika	4
USN-5	Created price per sq.ft & age calculations	Medium	Sai Charan	3

Sprint 2: Development & Documentation

Duration: 8 Feb – 15 Feb 2026 (8 Days) | **Total Story Points:** 16

Task ID	User Story / Task	Priority	Assigned To	Points
USN-6	Built histograms, bar charts & feature analysis	High	Chandrika	4
USN-7	Created interactive Tableau dashboard	High	Rammya Sre	4
USN-8	Analyzed renovation & feature impact trends	High	Narasimha	3
USN-9	Tested filters, performance & accuracy	High	Narasimha	3
USN-10	Prepared architecture & solution reports	Medium	Bhoomika	2

4. Performance Metrics

- **Total Points Completed:** 31 Story Points over two sprints.
- **Sprint 1 Velocity:** 2.14 points per day (15 points / 7 days).
- **Sprint 2 Velocity:** 2.00 points per day (16 points / 8 days).
- **Completion Rate:** Both Sprint 1 and Sprint 2 achieved a **100% completion rate** relative to planned points.

6. FUNCTIONAL AND PERFORMANCE TESTING

Data Integrity & Rendering

The core housing dataset, which includes property features such as bedrooms, bathrooms, floors, and renovation history, was successfully rendered in Tableau.

Preprocessing Steps Taken:

- Removed all duplicate records to ensure data accuracy.
- Handled missing values specifically within the renovation history fields.
- Standardized year fields into numeric formats for analytical compatibility.
- Validated overall data consistency and established realistic price ranges.

Analytical Logic (Calculated Fields)

The system uses custom logic to derive deeper insights beyond the raw data:

- **House Age:** Calculated as Current Year – Year Built.
- **Renovation Metrics:** Tracking "Years Since Renovation" to determine market impact.
- **Pricing Benchmarks:** Includes Price per Square Foot, Average Sale Price, and Median Sale Price.

Visualization & Functional Design

The dashboard environment is designed for high interactivity and multi-dimensional analysis.

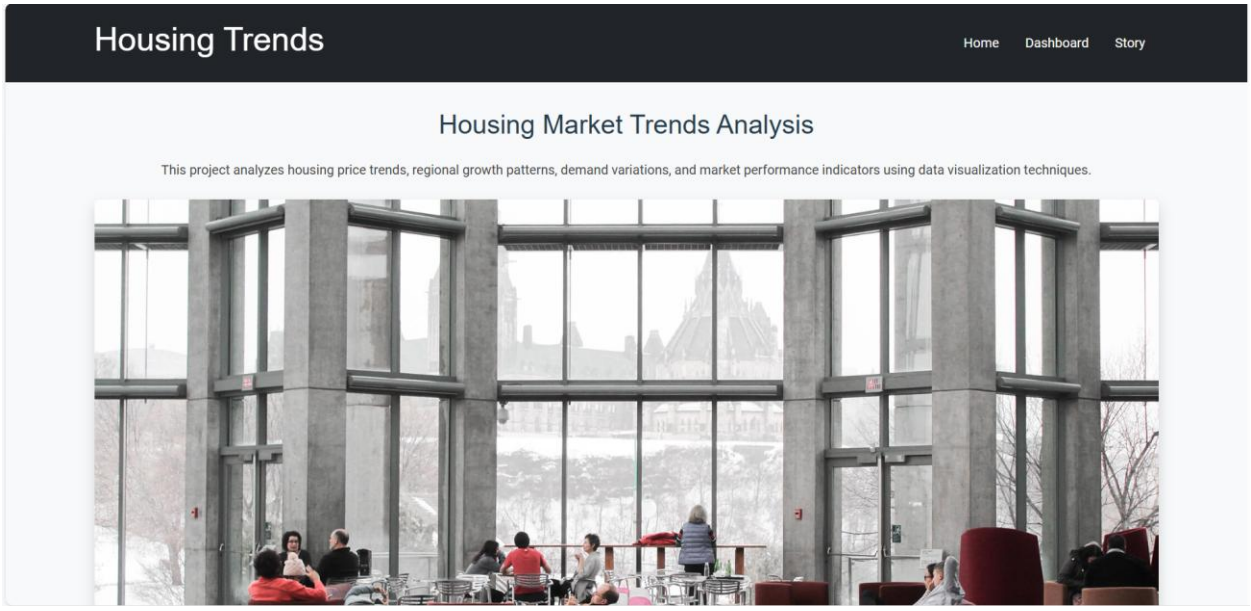
- **Filter Utilization:** Interactive filters are implemented for **Sale Price**, **Sale Price (bin)**, **Measure Names**, and **Age of House (in Years)**.
- **Dashboard Composition:** The final deliverable consists of **4 individual sheets** integrated into **1 comprehensive dashboard**.
- **Key Visualizations:**
    - **Overall Data Overview:** High-level summary of total area and average pricing.
    - **Total Sales by Years Since Renovation:** Histogram showing volume trends based on property updates.
    - **Distribution of House Age by Renovation Status:** Pie chart identifying the age groups of renovated vs. non-renovated homes.
    - **Feature Distribution:** Bar charts comparing No. of Bathrooms, Bedrooms, and Floors across various house ages.

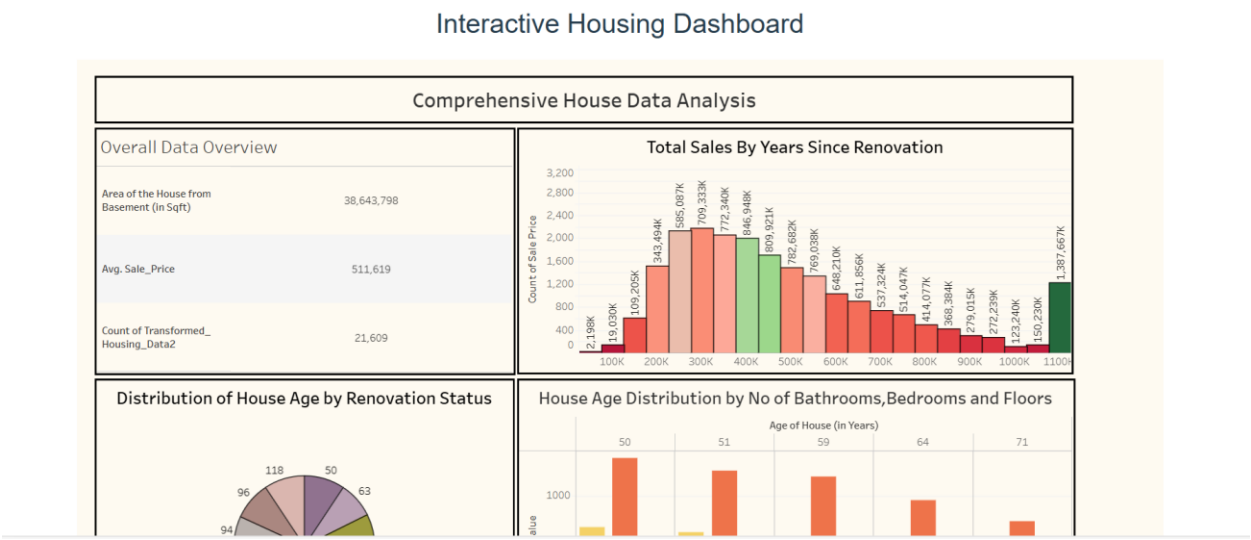
Performance Testing Results

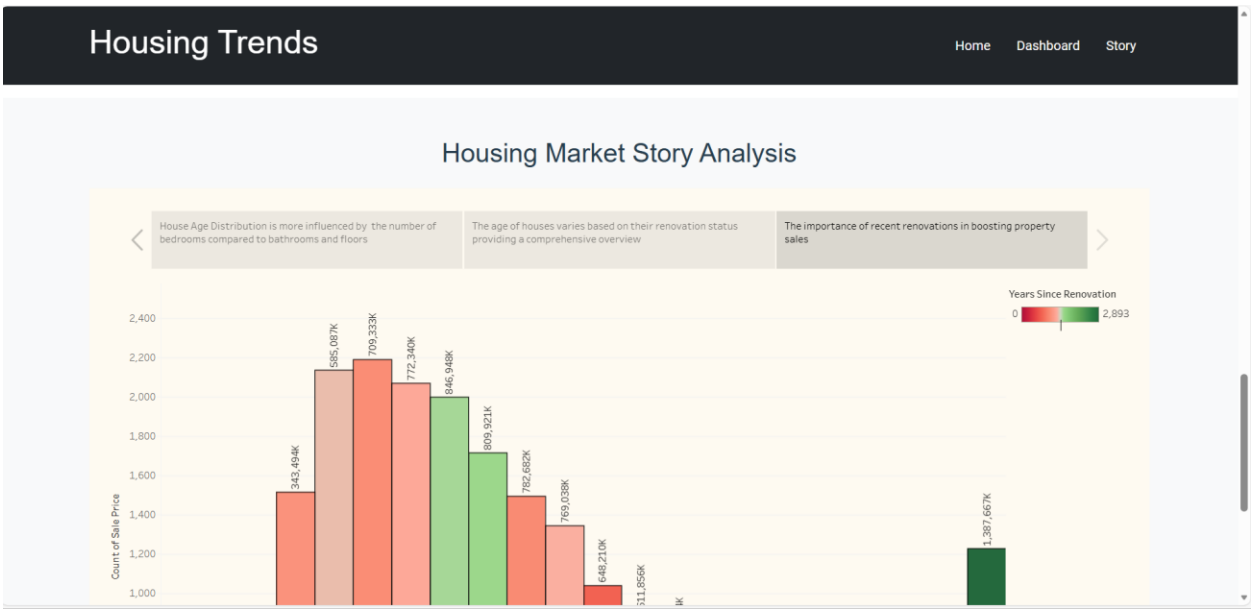
- **Functional Success:** The system successfully filters data dynamically without loss of accuracy.
- **Insight Delivery:** Story Design features are used to highlight that house age distribution is more heavily influenced by the number of bedrooms compared to bathrooms or floors.
- **User Experience:** The "Story" mode provides a guided narrative on the importance of recent renovations in boosting property sales

7. RESULTS

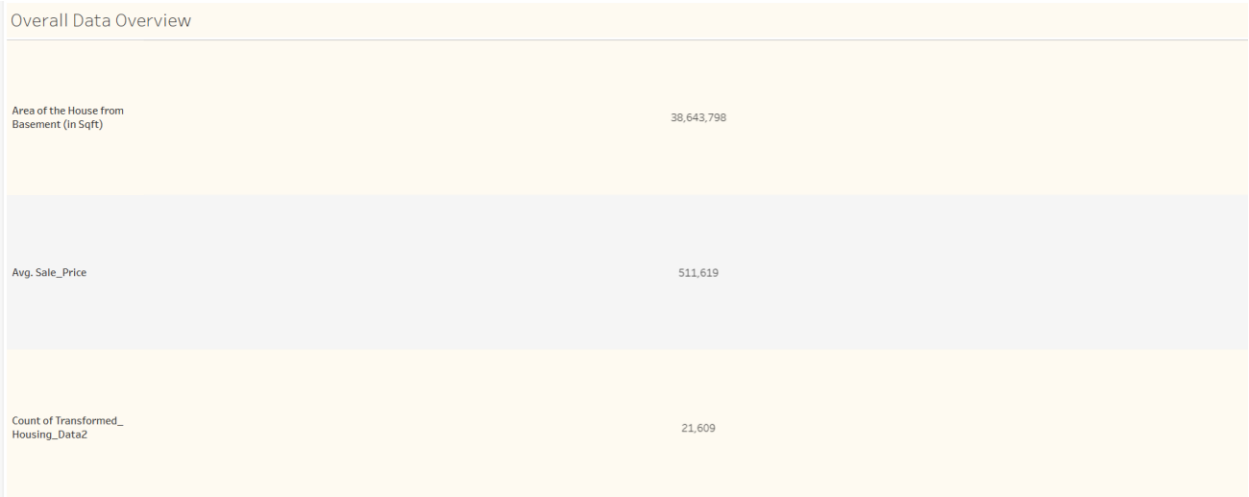
7.1 Output Screenshots





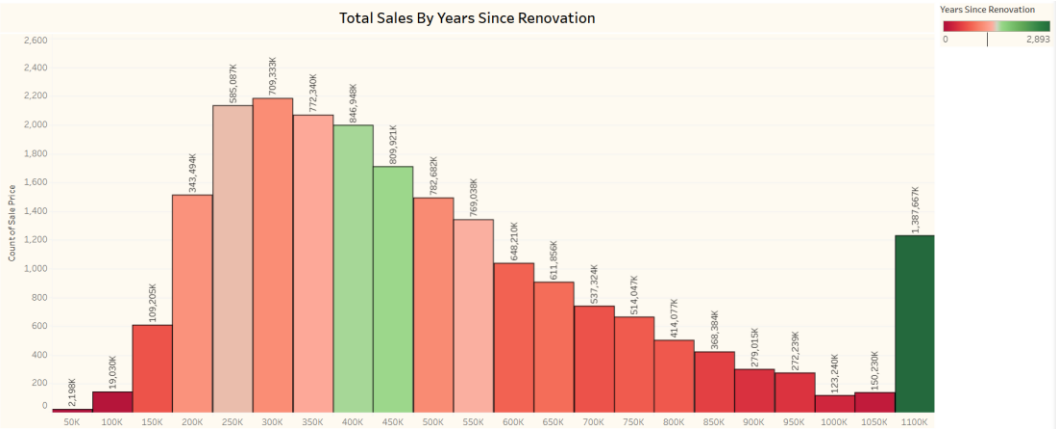


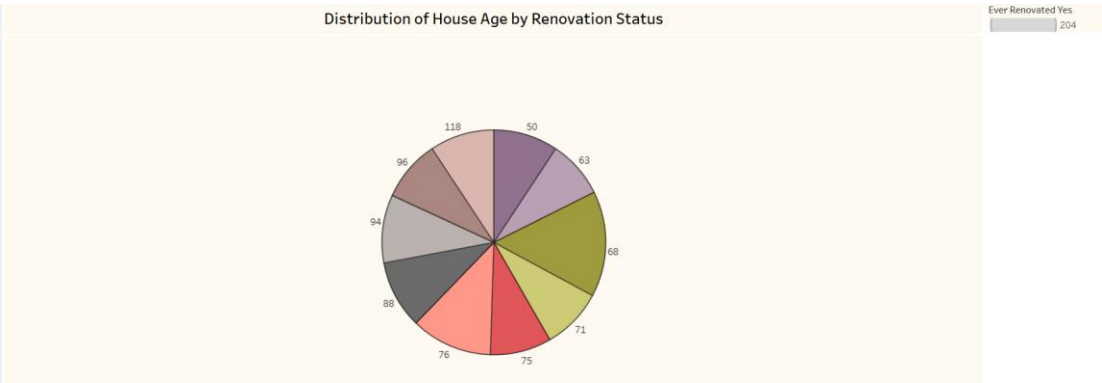
SHEETS:

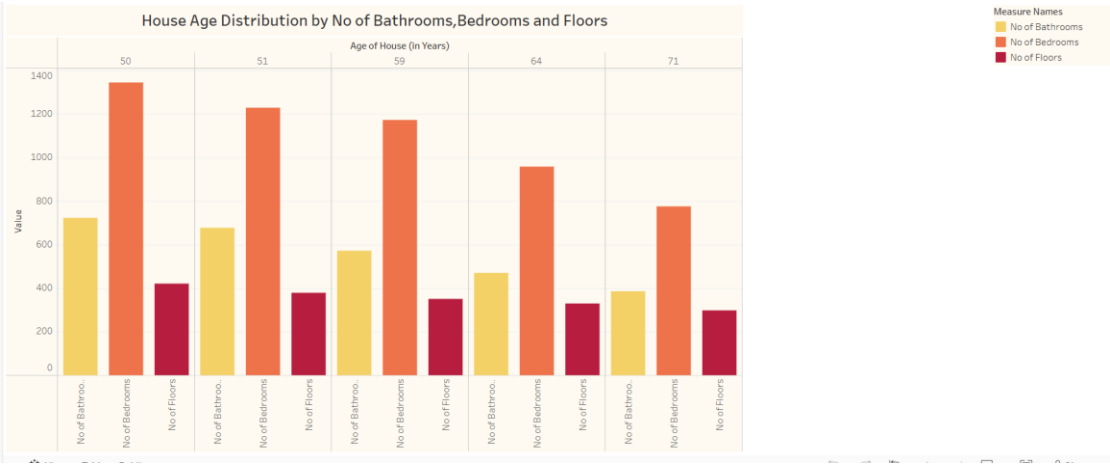


Overall Data Overview

Area of the House from Basement (In Sqft)	38,643,798
Avg. Sale_Price	511,619
Count of Transformed_ Housing_Data2	21,609




Distribution of House Age by Renovation Status
Ever Renovated Yes
204
118
50
63
96
94
68
88
71
76
75


House Age Distribution by No of Bathrooms,Bedrooms and Floors
Age of House (in Years)
50
51
59
64
71
Value
1400
1200
1000
800
600
400
200
0
No of Bathroo..
No of Bedrooms
No of Floors
Measure Names
No of Bathrooms
No of Bedrooms
No of Floors

8. ADVANTAGES & DISADVANTAGES

ADVANTAGES

- **Unified Data Visualization**: Consolidates multi-dimensional housing attributes (age, renovation, and structural features) into a single integrated view, replacing scattered and isolated data analysis.
- **Enhanced Decision-Making**: Provides stakeholders with actionable pricing insights and transparent, data-backed strategies, which improves confidence among analysts and executives.
- **Interactive Insights**: Unlike static reporting, the dashboard utilizes dynamic filtering (age, bedrooms, floors) and storytelling views to highlight specific pricing drivers.
- **Operational Efficiency**: Automates the preprocessing of raw data, including cleaning, handling missing values, and performing complex calculations like price per square foot.
- **Reduced Pricing Bias**: Promotes more objective property valuations by identifying the true relationships between property features and market value.
- **Scalability**: The architecture is designed to handle larger datasets and can be expanded to include regional or national real estate indicators.

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8. DISADVANTAGES

- **Data Dependency**: The accuracy of the dashboard's insights is entirely dependent on the quality and consistency of the input .csv housing datasets.
- **Learning Curve for Stakeholders**: While designed for usability, non-technical users may initially require guidance to fully utilize advanced drill-down views and story-based visuals.
- **Maintenance Requirements**: Regular updates are necessary to ensure the dashboard reflects evolving market trends and to replace or update housing datasets as they become obsolete.
- **Infrastructure Costs**: Operating as a subscription-based or enterprise BI solution may involve ongoing costs related to cloud hosting (AWS) and Tableau licensing.
- **Internet Dependency**: As a cloud-hosted Tableau solution, the dashboard rquires consistent internet connectivity for real-time access and interactive filtering.

9.CONCLUSION

The **Housing Market Trends & Pricing Analysis Dashboard** successfully bridges the gap between complex, multi-dimensional housing data and strategic decision-making. By transitioning from static, isolated spreadsheets to a unified **Tableau-based analytical platform**, this project enables real-time exploration of property value drivers.

Key achievements of the project include:

- **Enhanced Data Clarity**: The system simplifies scattered datasets, providing analysts with a clear visualization of how house age and features influence pricing trends.
- **Evidence-Based Insights**: Through calculated metrics like **Price per Square Foot** and **Years Since Renovation**, stakeholders can now identify hidden value patterns with higher statistical confidence.
- **Strategic Empowerment**: The interactive nature of the dashboard reduces pricing bias and uncertainty, allowing marketing and executive teams to position properties more competitively.
- **Scalable Infrastructure**: The integration of **Python** for robust data preprocessing and **AWS** for cloud storage ensures the solution is reliable and capable of scaling to national-level datasets in the future.

In summary, this project provides a high-impact, data-driven solution that enhances the efficiency and accuracy of real estate market analysis, ensuring that stakeholders can make informed investment decisions in an increasingly data-centric industry.

10. FUTURE SCOPE

The current analytical system establishes a strong foundation for data-driven real estate insights, but it is designed with high scalability to evolve alongside emerging technologies. Future enhancements include:

- **Integration of Predictive Analytics**: Transitioning from descriptive to predictive modeling by incorporating **Machine Learning (ML) algorithms** to forecast future property price appreciation based on historical trends.
- **Real-Time Data Integration**: Connecting the dashboard to **live Real Estate APIs** to provide up-to-the-minute market indicators and price fluctuations rather than relying on static .csv uploads.
- **Geospatial & Neighborhood Mapping**: Expanding the visualization layer to include **GIS-based maps** that analyze the impact of neighborhood-specific variables such as proximity to schools, public transport, and crime rates.
- **Automated Pricing Recommendations**: Developing a module that suggests optimal listing prices for new properties by automatically comparing them to similar historical "feature combinations" within the dataset.
- **Mobile Accessibility**: Developing a mobile-responsive version of the dashboard or a dedicated app to allow property investors and agents to access insights on-the-go.
- **Regional & National Scaling**: Expanding the scope of the data processing engine to handle massive datasets covering multiple cities or entire countries for broader market comparisons.